



A robust deep learning based model for denoising phonocardiogram signals in clinical environments

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ABSTRACT

A cardiac auscultation, being a very popular first line cardiovascular diseases screening method, typically produces phonocardiogram signals contaminated by background sounds, e.g., speech and movement of other patients and doctors/nurses, phone ringing and door knocking/opening/closing, and internal physiological body noises, e.g., lung sounds, muscle contraction and motion artifacts, as it is usually done in a noisy environment and it is not possible to suppress all the internal physiological body noises. So, a quality of the recording is finally decreased. Moreover, this can strongly affect a subsequent analysis/classification of the captured recording and can also lead to its incorrect classification. In this paper, we propose a novel deep convolutional architecture that integrates the bidirectional long short-term memory layers with the transformer blocks involving multi-head attention mechanism, which is particularly designed for denoising phonocardiogram signals. The performance of the proposed artificial intelligence-based approach is evaluated through synthetic data generated by using one public phonocardiogram dataset and two public real-world hospital ambient/respiratory sound databases as well as by deploying two public and renowned real-world clinical environment datasets containing phonocardiogram signals contaminated by various environmental and physiological noises recorded during the clinical auscultations. The deployed complex evaluation strategy assures a higher robustness and generalizability of the proposed approach. As also a result of that, the proposed model outperforms the current state-of-the-art approaches and provides the significant improvements in all the assessed signal quality and noise suppression metrics and over all the involved benchmark datasets. Finally, it should be noted here that the model achieved an average estimated SNR of 14.984 dB and 11.924 dB for the PASCAL and CirCor22 dataset respectively, which represent the renowned real-world clinical environment datasets.

1. Introduction

Cardiovascular diseases (CVDs) symbolize the leading cause of mortality and morbidity worldwide, causing 300,000 deaths in North America annually (Chan et al., 2019) and approximately one-third of deaths globally (Chen et al., 2021; Samanta et al., 2019). 85% of those deaths are typically caused by a heart attack and stroke (Chen et al., 2021). CVDs have also a serious impact on a financial situation of persons suffering from this kind of diseases as well as their families, especially in low- and middle-income countries (Miranda et al., 2019). Therefore, early diagnosis and treatment are of paramount importance when it comes to lessening the adverse outcomes of CVDs and reducing a mortality rate. Moreover, telemonitoring approaches can provide better management of chronic CVDs, transforming their treatment from a reactive to a proactive approach (Gómez-García et al., 2021) and by doing so improving a quality of medical services and reducing expenses

in this case. To be more concrete, a decision support system with alert feedback, which is implemented in a telemonitoring approach, can help physicians to achieve an early diagnosis, management and treatment of CVDs. It is also worth noting here that the above-specified telemonitoring approaches are also highly relevant in the context of Active and Assisted Living (AAL) concept (Despotovic et al., 2022), which covers a whole range of intelligent and adaptive assistive technologies that allows an effective, ubiquitous and unobtrusive support for an elderly, impaired and frail population, thus trying to address the major health and societal challenges and demographic changes nowadays. So, an AAL health screening system is in principle supposed to monitor a health data, to allow an early diagnose of a disease, for instance CVDs, or track its progress, enabling better patient care, well-being and improved quality of life, but also making more optimal

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use of healthcare resources by reducing hospital admissions and costly interventions.

Cardiac auscultation represents a non-invasive and simple, yet useful first line CVDs screening method, which is more than 180 years old (Liu et al., 2016) and enables a detection of the structural heart abnormalities or evaluating them in a differential diagnosis. It is very essential for an early diagnosis of CVDs due to its good performance for reflecting a cardiovascular system, to be more precise a mechanical motion of the heart, and its noninvasiveness (Chen et al., 2021). Anyhow, substantial clinical experience and skilled physicians are still required in this case (Mangione and Nieman, 1997). Moreover, it should be noted here that some heart sounds are in the frequency range, which is inaudible for humans (Chen et al., 2021). Therefore, computer-assisted decision support systems based on auscultation can be used to assist a timely diagnosis and treatment of CVD patients (Clifford et al., 2017; Reyna et al., 2022; Han et al., 2024). Their application in a daily practice can bring a substantial improvements to this area of human health management (Chen et al., 2021).

A digital stethoscope represents a non-invasive acoustic diagnostic instrument, which is typically deployed to record heart sounds as phonocardiogram (PCG) signals during cardiac auscultation. It captures the acoustic vibrations made by the heart with a microphone-transducer. When it comes to more information about the digital stethoscope, the interested reader is referred to Swarup and Makaryus (2018). An introduction of wearable PCG sensors (Lin et al., 2021), which have a huge potential for in-home cardiac care, made a real-time remote CVDs screening feasible. It should be highlighted here that an in-home environment, which is highly relevant in the context of the telemonitoring approaches as well as the AAL concept already mentioned above, is more challenging than a clinical environment for heart sounds' acquisition, given the fact that unskilled/naive users mostly control the PCG sensors. Ambient artifacts, e.g., background sounds (speech and movement of other patients and doctors/nurses, phone ringing and door knocking/opening/closing) and power supply interference, and internal physiological body noises, e.g., lung sounds, muscle contraction and motion artifacts, contaminate a heart sound captured in a clinical environment (Altuve et al., 2020), which becomes even more noisy when it is recorded in in-home or AAL scenarios. So, a quality of PCG recording is finally reduced and a performance of machine learning based automated CVDs diagnostic procedures is thus strongly affected (Gradolewski and Redlarski, 2014; Roy et al., 2023). Therefore, reducing an amount of noise included in PCG signal captured during cardiac auscultation is vital to ensure the reliability, robustness and clinical interpretability of the models developed for screening of CVDs.

Our contribution in this paper, strongly motivated by the remarkable achievements obtained by the machine learning based approaches in the related research fields, e.g., heart sound classification/cardiovascular disease detection (Shaker et al., 2024; Chen et al., 2024; Gao et al., 2025), automatic speech recognition (e.g., wav2vec 2.0 (Baeveski et al., 2020) and Whisper (Radford et al., 2022)) or speech and language technologies (i.e., the large language models used in natural language processing, e.g., LLaMA2 (Touvron et al., 2023) and GPT4 (OpenAI, 2023)), as well as a scarcity of the machine learning based heart sound denoising approaches published in the literature, see Section 2 for more details, and very promising results published in Ali et al. (2023), which were achieved by a deep learning approach, is two-fold:

1. We proposed a deep convolutional architecture entitled T-BiLSTM that integrates bidirectional long short-term memory (BiLSTM) layers with transformer blocks involving a multi-head attention mechanism (MHA), which was particularly designed for denoising PCG signals. The proposed T-BiLSTM framework/architecture leverages the strengths of sequential learning and attention-driven feature extraction, achieving a superior

performance in a PCG signal denoising under the challenging clinical conditions. In comparison to the previous methods published in the literature, a novelty of our approach lies in embedding both the BiLSTM and transformer components directly into each encoder stage of the model, enabling simultaneous learning of temporal dynamics and global dependencies in a unified architecture.

2. Unlike the previous studies, e.g., Ali et al. (2023) and Yang et al. (2023), that rely on rather limited and small PCG dataset/s, our approach incorporates five diverse and rather big PCG datasets coming from the different sources, which are supposed to enhance a generalizability and robustness of the proposed denoising model. It is worth noting here that the deployed multi-dataset training strategy allows the model to adapt to a broader spectrum of recording settings, clinical scenarios and noise patterns in its training phase. Moreover, we evaluated the model's performance on real noisy recordings, highlighting its robustness under realistic auscultation conditions (see Fig. 1).

The remainder of the paper is organized as follows. Section 2 describes the related work in the corresponding research area, i.e., heart sound denoising/noise reduction. The proposed approach is, in detail, presented in Section 3. Section 4 provides a detailed description of all the experiments done in this study. The experimental results, to be more precise the results obtained for a benchmark and ablation study, are presented in Section 5 and discussed in Section 6. The conclusion and possible future work are given in Section 7.

2. Related work

The studies proposing the heart sound denoising and noise reduction methods/approaches can be divided roughly into two groups, i.e., the papers presenting the methods/approaches based on traditional signal processing techniques, e.g., mean square algorithms, wavelet transform, Bayesian techniques, Kalman filter and adaptive filtering, and the papers proposing the machine learning based methods/approaches. It is worth noting here that the first group considerably dominates the current literature, probably due to a lack of publicly available datasets, which are critically needed for a development of machine learning based methods/approaches.

The approaches based on traditional signal processing techniques are described in the following papers (Paul et al., 2006; Almasi et al., 2013; Gradolewski and Redlarski, 2014; Cheng and Zhang, 2014; Jain and Tiwari, 2016; Ali et al., 2017; Zhang et al., 2022; Minani and Malarvili, 2023; Pham et al., 2018; Nazemi et al., 2021; Grooby et al., 2021, 2022; Pauline and Dhanalakshmi, 2022; Shervegar, 2022, 2025; Reddy et al., 2023; Brindha and Manjula, 2024). In Paul et al. (2006), the authors designed the noise reduction method founded on spectral domain minimum-mean squared error estimation originally introduced in Ephraim and Malah (1984), which was further enhanced, in the paper, by an electrocardiography gating, soft thresholding rule, iterative residual noise estimation and forward-backward noise estimation in order to remove, in effective manner, diverse interfering noises, i.e., lung sounds, muscle contraction, and background noise, while preserving critical low amplitude heart sounds. A model-based Bayesian denoising framework, which is based on a novel dynamical model for PCG allowing a realistic synthesis of PCG signals and the extended Kalman smoother, being deployed as the Bayesian filter, was described in Almasi et al. (2013). Wavelet transform based family of the denoising approaches is represented by the following papers (Gradolewski and Redlarski, 2014; Cheng and Zhang, 2014; Jain and Tiwari, 2016; Ali et al., 2017; Zhang et al., 2022; Minani and Malarvili, 2023). In Pham et al. (2018), the authors developed a novel PCG noise reduction technique based on nonnegative matrix factorization of the PCG spectrogram and adaptive contour representation computed from the PCG short-time Fourier transform. An auto-regressive

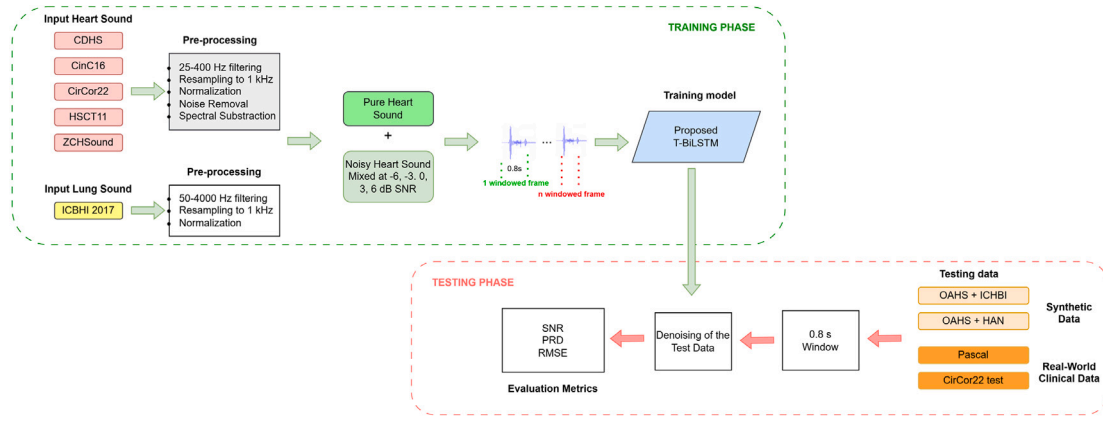


Fig. 1. A graphical representation of the proposed denoising system. Heart sound and lung sound recordings undergo preprocessing, followed by mixing at specific SNR levels to generate noisy heart sound data. During training, 0.8 s long windows are extracted and processed by the proposed T-BiLSTM model. In the testing phase, the trained model is applied to unseen noisy datasets, including both synthetic and real-world clinical data, and evaluated using the standard denoising metrics.

model based Kalman filter was proposed in Nazemi et al. (2021) to reduce the noises in PCG signal. In Grooby et al. (2021), the authors present a novel non-negative matrix co-factorization based approach in order to acquire high quality heart and lung sounds from noisy PCG recordings, specifically for the newborn population. An improved version of this approach is described in Grooby et al. (2022). An optimal and robust low-cost adaptive filter system to estimate a noise-free signal with high accuracy using the least mean square algorithm was specifically designed in Pauline and Dhanalakshmi (2022) for PCG denoising. Moreover, an optimal sliding window adaptive filter system for denoising PCG signals was proposed in Shervegar (2022). A robust version of this system is published in Shervegar (2025). The authors in Reddy et al. (2023) developed a new noise reduction approach founded on fast iterative filtering for PCG based cardiac diseases computer-aided diagnosis. Finally, an empirical mode decomposition based PCG denoising approach is presented in Brindha and Manjula (2024).

When it comes to the machine learning based methods/approaches, which has been mushrooming just in the last two years, they are represented by the following studies (Ali et al., 2023; Yang et al., 2023; Al-Zaben et al., 2024; Guo and Tang, 2024). In Ali et al. (2023), the authors proposed a first deep learning framework termed LU-Net specifically designed for suppressing surrounding and internal lung noises in noisy PCG recordings. The authors in Yang et al. (2023) developed a two stage noise reduction approach based on a conventional adaptive noise cancellation method (Ramli et al., 2012) for a course denoising and deep neural network (DNN) entitled the dual-channel interactive noise cancellation network (DINCNet) for further improvement. A novel approach deploying a modified cosine transform and long short-term memory based masking to effectively differentiate heart sounds from interfering sounds and background noise was described in Al-Zaben et al. (2024). Finally, the authors in Guo and Tang (2024) designed a PCG denoising method for wearable multi-channel heart sound signals, which is based on an independent component analysis and applies a convolutional neural network for group classification purposes. It should be noted here that it requires an ECG signal, which is used as a time marker in this method.

3. Proposed approach

3.1. Problem formulation

Cardiac auscultation plays a pivotal role in the early detection and diagnosis of cardiovascular diseases. However, its efficacy is often compromised by extraneous noise arising from respiratory sounds, hospital environments, and mechanical interference. These unwanted components obscure critical heart sound features, increasing the difficulty of

manual interpretation and reducing diagnostic reliability. The primary challenge, therefore, is to extract diagnostically relevant PCG signals from real-world noisy recordings while preserving their physiological integrity. Mathematically, the recorded PCG signal can be represented as:

$$y = x + n, \quad (1)$$

where $y \in \mathbb{R}^{1 \times N}$ denotes the acquired noisy PCG signal, $x \in \mathbb{R}^{1 \times N}$ represents the clean heart sound, and $n \in \mathbb{R}^{1 \times N}$ corresponds to the additive noise that degrades the signal quality. The objective is to estimate the clean heart sound $\hat{x}$ from y by learning a nonlinear mapping function $F(\cdot)$:

$$\hat{x} = F(y; \theta) \quad (2)$$

where θ represents the trainable parameters of the model. The function $F(\cdot)$ must effectively learn complex signal T-BiLSTM formations that suppress noise while preserving the underlying cardiac sound characteristics.

To achieve robust PCG denoising, we adopt a deep learning-based end-to-end framework capable of dynamically adapting to complex noise distributions and optimizing a feature extraction. Unlike conventional filtering techniques that rely on predefined signal processing methods, our model is trained to minimize a reconstruction loss function that quantifies a discrepancy between an estimated clean signal and its ground truth counterpart. Specifically, a denoising process is optimized by minimizing a mean squared error (MSE) loss between an estimated clean signal $\hat{x}_i$ and its ground truth counterpart x_i :

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N \|\hat{x}_i - x_i\|^2, \quad (3)$$

where N denotes the number of training samples. The model parameters θ are iteratively optimized using the Adam optimizer, ensuring stable convergence and robust performance across diverse real-world PCG recordings.

3.2. Proposed model architecture

The proposed T-BiLSTM network is a novel transformer-enhanced BiLSTM-based encoder-decoder framework designed for heart sound denoising (see Fig. 2). The model synergistically integrates convolutional feature extraction, bidirectional recurrent sequence modeling, and self-attention mechanism to enhance noise suppression while preserving critical PCG signal patterns. By incorporating MHA within a hierarchical encoding structure, the model captures both short-range

and long-range dependencies, ensuring that diagnostically relevant acoustic features are retained.

The encoder consists of five encoder blocks, each denoted as Encoder_i (where $i = 1$ to 5). The first block, Encoder_1 , processes the raw noisy heart sound input x_i using a 1D convolutional operation with kernel size $K = 31$, stride $S = 1$, and ReLU activation. The choice of kernel size $K = 31$ is motivated by the recent findings that claims that larger convolutional kernels can enhance the model's capacity to capture long-range dependencies in time series. Specifically, Ding et al. (2022) demonstrated that kernels of size 31 can approximate the global receptive field benefits typically seen in attention-based models, while retaining the computational advantages of convolution. For heart sound processing, this kernel size corresponds to a ~ 32 ms window at 1 kHz sampling rate, which aligns with the duration of key acoustic events such as S1 and S2. The first event/sound is produced by the closure of the atrioventricular valves (tricuspid and mitral) at the beginning of systole, called a ventricular contraction in the literature. The second one is caused by the closure of the semilunar valves (pulmonary and aortic) at the beginning of diastole, entitled a ventricular relaxation in the literature.

In the convolutional layers of the following four encoder blocks, a stride of 2 reduces the temporal resolution of the output feature maps, downsampling the temporal dimension along the encoding path. Concurrently, a number of channels increases to 32, 32, 64, and 64 for the respective blocks, enhancing a feature representation capacity.

Each convolutional output is processed by a BiLSTM layer. The BiLSTM layers contain 8, 16, 16, 32, 32 hidden units across five encoder blocks and are initialized using He-normal weight distribution. These BiLSTM layers improve sequential feature retention by aggregating information from both past and future time steps, mitigating temporal distortions introduced by noise. It should be noted here that the BiLSTM is a type of recurrent neural network that processes an input sequence in both forward and backward directions. This enables the model to access both past and future context at each time step, which is particularly beneficial for tasks like denoising where temporal dependencies are distributed across the entire signal. The convolutional layer following the final encoder block, Encoder_5 , produces a compressed latent representation of the noisy input, which serves as the model's bottleneck.

To overcome the inherent limitations of recurrent networks in modeling long-range dependencies, the proposed T-BiLSTM model integrates transformer blocks with MHA within the encoding pathway, where one transformer block is placed after the second one and each subsequent BiLSTM layer to enhance the modeling of contextual dependencies. The transformer component significantly improves the model's ability to distinguish between heart sound components and noise artifacts by dynamically assigning attention weights to different temporal segments. The core of the transformer mechanism is the scaled dot-product attention, computed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (4)$$

where Q , K , and V represent the query, key, and value matrices, and d_k is the dimensionality of the key vectors. By using multiple attention heads ($\text{num}_h = 8$), the model captures diverse feature representations, allowing it to focus on different temporal segments of the heart sound simultaneously (see Fig. 3).

Each MHA block is followed by a feed-forward network (FFN), defined as:

$$\text{FFN}(x) = \text{ReLU}(W_1 x + b_1)W_2 + b_2, \quad (5)$$

where W_1 and W_2 are learnable weight matrices. The inclusion of residual connections ensures stable gradient propagation, preventing vanishing gradient issues during training.

By combining BiLSTM and MHA, the model leverages both sequential memory and context-aware attention, allowing it to enhance signal

reconstruction while filtering out stochastic noise components. Unlike traditional RNN-based architectures, which struggle with long-range dependencies, the integration of transformer-based attention improves signal differentiation, making it robust to varying noise conditions in real-world auscultation environments.

The decoder is structured symmetrically to the encoder and is responsible for reconstructing the clean heart sound while retaining diagnostically significant temporal structures. The first decoder layer (in Decoder_5) takes the bottleneck feature representation and applies a convolutional transformation to restore dimensionality. Each decoder block, Decoder_i , utilizes 1D convolutional layers with kernel size $K = 31$, followed by ReLU activation and UpSampling1D operations to progressively recover the original temporal resolution.

To ensure efficient signal reconstruction, skip connections link each Encoder_i with its corresponding Decoder_i , reinforcing a retention of fine-grained signal details. Each decoder block after Decoder_5 receives the concatenated output features from the previous block and transformer_{i+1} , enabling a context-aware refinement of the reconstructed output. Skip connections facilitate a feature reusability while enhancing a stability of model training and convergence (Li et al., 2018). The final decoding layer applies a 1D convolutional operation with a hyperbolic tangent (tanh) activation to generate the clean heart sound waveform, ensuring amplitude consistency with real-world physiological signals. The code is available at https://github.com/MLForge-M/PCG_Denoising_T-BiLSTM.

3.3. Benchmark systems

To evaluate the proposed denoising framework, we selected three widely used deep learning architectures: fully convolutional networks (FCN), U-Net, and LU-Net. These models, commonly applied in biomedical signal processing, serve as benchmarks to assess noise suppression and signal reconstruction. Each architecture is designed to process 1D audio signals, leveraging convolutional and recurrent structures to enhance feature extraction and noise filtering. Each model is described separately in the following paragraphs summarizing its key technical specifications.

A. Fully Convolutional Network (FCN)

FCN used in this study is inspired by the architecture proposed in Chiang et al. (2019), with its structure adapted for heart sound denoising. Designed as a deep denoising autoencoder (DAE), the model follows an encoder-decoder framework, where the encoder progressively compresses the input signal, and the decoder reconstructs the clean waveform. The encoder consists of stacked convolutional layers with batch normalization and exponential linear unit (ELU) activations, reducing the input heart sound signal into a compact feature space. The first convolutional layer applies 40 filters of size 16×1 with a stride of 2, followed by three layers with 20 filters of the same kernel size and stride. A final convolutional layer with a single filter extracts the latent representation, which is 32 times smaller than the original input. The decoder, mirroring the encoder, employs deconvolutional layers to restore the temporal resolution while preserving structural details. The final layer applies a deconvolution operation with a single filter to generate the denoised output. While FCN is computationally efficient, its lack of explicit long-term dependency modeling may limit its ability to preserve fine-grained temporal details in heart sound recordings.

B. U-net

U-Net (Ronneberger et al., 2015) follows an encoder-decoder framework with skip connections that facilitate information retention across layers. The encoder is composed of five convolutional blocks, each containing two 1D convolutional layers with a kernel size of 31 and ReLU activation for non-linearity. The convolutional layers with a stride of 2 are employed for progressively reducing the signal resolution while increasing the receptive field. The bottleneck layer incorporates dilated

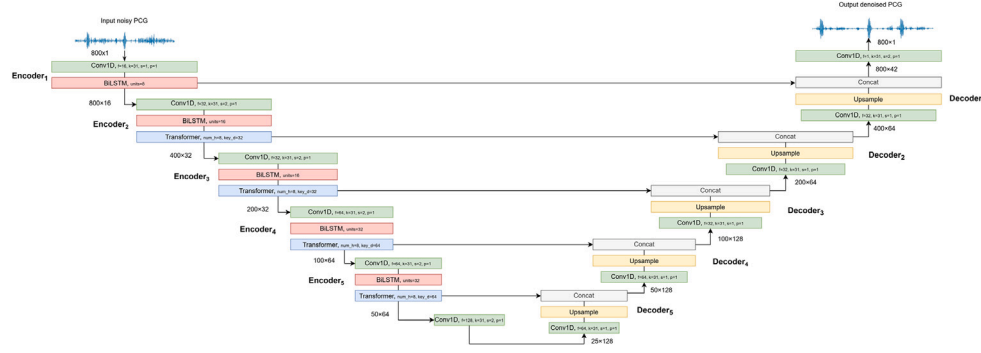


Fig. 2. Block diagram of the proposed T-BiLSTM architecture for PCG denoising. The model follows an encoder–decoder structure, starting from the noisy input (top left) and ending up with the denoised output (top right). The encoder includes Conv1D, BiLSTM, and transformer layers with downsampling, while the decoder applies upsampling and skip connections. Layer configurations such as a filter size and stride are shown at each stage.

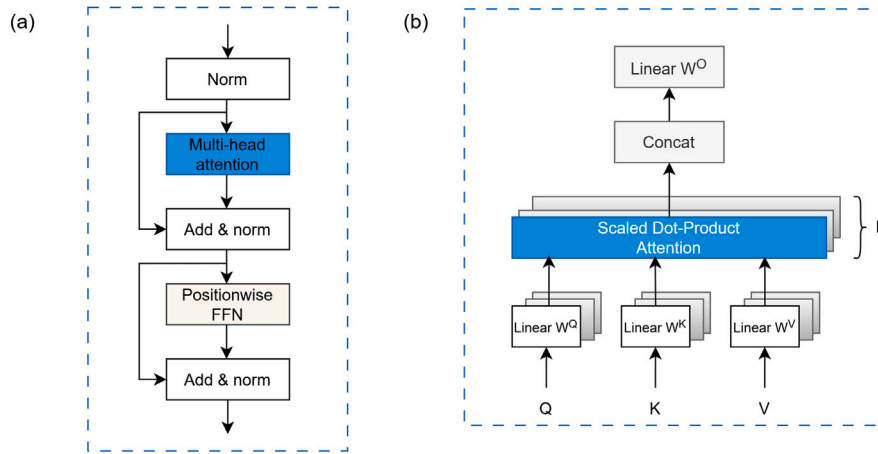


Fig. 3. Structure of the transformer block used in the T-BiLSTM architecture. (a) transformer encoder layer composed of MHA, positionwise FFN, layer normalization, and residual connections. (b) MHA, where query (Q), key (K), and value (V) inputs are linearly projected, processed via scaled dot-product attention, concatenated, and passed through a final linear layer.

convolutions to enhance feature extraction and maintain contextual information across the sequence. The decoder mirrors the encoder, employing upsampling to restore the original resolution. Skip connections link corresponding encoder and decoder layers, ensuring the preservation of high-resolution details throughout the reconstruction process. While U-Net effectively balances feature extraction and signal reconstruction, its relatively small bottleneck receptive field can limit its ability to capture long-range dependencies in sequential data.

C. LU-net

LU-Net (Ali et al., 2023) is an advanced variant of U-Net that integrates BiLSTM modules into skip connections to enhance temporal feature retention. The LU-Net architecture consists of five encoder layers, each comprising a 1D convolutional layer with a kernel size of 31, batch normalization for stable training, and ReLU activation. A compression path reduces dimensionality using convolutional layers with a stride of 2, creating a compact representation at the bottleneck. BiLSTM modules are integrated into the skip connections to capture forward and backward dependencies within the signal, enhancing temporal coherence. The decoder follows an upsampling strategy with transposed convolutions and ReLU activation, gradually restoring the signal resolution. He-normal kernel initialization and hyperbolic tangent (tanh) activation are employed to stabilize gradient propagation during training. The incorporation of BiLSTMs enables LU-Net to preserve long-range dependencies, making it particularly effective for noise suppression in sequential biomedical signals.

4. Experiments

To evaluate the effectiveness of the proposed denoising framework, we conducted series of controlled experiments using diverse heart sound datasets. The experimental protocol was designed to ensure consistency in data preprocessing, model training, and evaluation. Importantly, exactly the same experimental setup was also applied in the ablation study to ensure a fair comparison across all the model variants. This includes a deployment of the identical training data, preprocessing steps, and evaluation metrics for every architectural configuration. The following subsections provide a detailed account of the datasets utilized in this study, preprocessing steps, experimental configuration, and performance evaluation criteria.

4.1. Datasets

This study utilizes two primary categories of datasets: heart sound datasets and real-world clinical environment datasets. The heart sound datasets comprise PCG recordings used for classification, segmentation, and disease detection. These datasets include clinically acquired heart sounds and curated repositories that provide benchmarks for automated auscultation systems. Key characteristics of the deployed heart sound datasets are presented in Table 1. The following heart sound datasets are used in this study:

1. **Cardiac Disease Heart Sound (CDHS)** (Tang et al., 2021) is a clinically acquired heart sound dataset consisting of 3875 PCG

Table 1

The deployed heart sound datasets and their key characteristics.

	CDHS	CinC16	CirCor22	HSCT11	OAHS	Pascal	ZCHSound
Num. participants	76	1297	1568	206	NA	NA	NA
Num. recordings	3875	2435	5282	412	1000	832	941
File length [s]	15–34	1–35.5	5.3–80.4	20–70	1.1–4.0	0.8–27.9	20
Total duration [h]	18.1	20.2	33.5	5.2	0.7	1.6	NA
Age categories	Adults	Mixed	Children	Adults	NA	Mixed	Children
Add. information	man. quality score	segmen.	segmen., socio-demo.	socio-demo.	NA	man. segmen.	NA
Samp. freq. [kHz]	1	2	4	11	8	4/44.1	8
Label categories	NA	2	2/3	Train - test match	NA	3/4	NA

recordings obtained from 76 patients. The recordings, ranging from 15 to 34 s. Each segment is labeled with a quality score from 1 (very poor) to 5 (excellent) for fidelity assessment.

2. **PhysioNet/Computing in Cardiology 2016 (CinC16)** (Clifford et al., 2016) aggregates nine independent heart sound datasets, comprising 2435 PCG recordings obtained from 1297 subjects. The dataset includes recordings from multiple auscultation sites (aortic, pulmonary, tricuspid, and mitral regions) with a sampling frequency of 2 kHz. It contains cases of heart valve diseases and coronary artery disease and serves as a benchmark for heart sound classification and segmentation tasks.
3. **CirCor DigiScope 2022 (CirCor22)** (Oliveira et al., 2021) is the most extensive publicly available pediatric heart sound dataset, collected in Northeast Brazil during pediatric screenings. It contains 5282 PCG recordings obtained from 1568 subjects. With a focus on early cardiac screening, the dataset includes recordings from healthy and pathological cases (1144 normal, 305 with cardiac pathologies, and 119 indiscernible), collected at 4 kHz with 16-bit resolution. The dataset is designed for murmur classification and clinical outcome prediction, featuring expert-annotated heart sound segmentations.
4. **Heart Sounds Catania 2011 (HSCT-11)** (Spadaccini and Beritelli, 2013) comprises PCG recordings obtained from 206 individuals (157 male, 49 female), recorded using the ThinkLabs Rhythm Digital Electronic Stethoscope at a sampling frequency of 11.025 kHz. The recordings were acquired at the pulmonary valve site under resting conditions, excluding post-exercise and pathological cases.
5. **The Open-Access Heart Sound(OAHS)** (Yaseen et al., 2018) compiles PCG recordings collected from 48 different sources, including medical literature and digital repositories. It includes 1000 recordings categorized into normal, aortic stenosis, mitral stenosis, mitral regurgitation and mitral valve prolapse classes. The dataset has an 8 kHz sampling rate, with each recording capturing exactly three heartbeats.
6. **Pascal dataset** (Gomes et al., 2013) was developed for the PAS-CAL 2011 Heart Sounds Challenge and includes PCG recordings obtained via digital stethoscopes and mobile phones in both clinical and non-clinical environments. The dataset supports heart sound segmentation (S1 and S2 detection) and classification (normal, murmur, extra heart sound, and artifacts).
7. **ZCHSound** (Jia et al., 2024) is a pediatric heart sound dataset comprising 941 recordings collected at three leading medical institutions in China. Data were acquired using the ChildCare G-100 smart stethoscope at 8 kHz, with manual annotations provided by expert cardiologists. The dataset includes high-fidelity samples and a subset reflecting real-world auscultation conditions.

The second category includes real-world ambient and respiratory sound datasets, which capture environmental noises and pathological respiratory sounds. These datasets provide realistic background conditions for evaluating automated heart sound processing systems to ensure their robustness against extraneous noises. The following real-world hospital ambient and respiratory sound datasets are used in this study:

1. **Hospital Ambient Noise (HAN)** (Ali and Shuvo, 2021) is a dataset of real-world audio recordings coming from hospital environments, comprising 562 audio segments with a length of 5 s. Extracted from a 68 min non-copyrighted hospital recording, the dataset captures common background sounds, including medical equipment beeps, patient conversations, and foot traffic. To align with a typical sampling rate of the PCG recordings, the authors applied a 3rd-order Butterworth low-pass filter with a 500 Hz cutoff frequency and resampled the data to 1000 Hz to match PCG recording standards.
2. **International Conference on Biomedical Health Informatics (ICBHI)** (Rocha et al., 2018) is a collection of 920 annotated respiratory sound recordings obtained from 126 subjects, totaling 5.5 h of audio. The dataset contains 6898 respiratory cycles, including 1864 with crackles, 886 with wheezes, and 506 exhibiting both. Sampling frequencies range from 4 kHz to 44.1 kHz, with recording durations between 10 and 90 s. Each recording is accompanied by an annotation file specifying respiratory cycle segmentation and the presence of adventitious sounds.

4.2. Data preparation

A. Training data

For training, we used heart sound recordings from the CDHS, CinC16, CirCor22, HSCT-11, and ZCHSound datasets. These datasets provide a diverse collection of PCG recordings, helping the model to generalize for different heart conditions and recording environments. A total of 53,913 recordings were used during training of the T-BiLSTM model.

All recordings were resampled to 1 kHz to standardize the data, ensuring consistency across the different sources. A third-order Butterworth bandpass filter with a passband of 25–400 Hz was applied to retain only the essential frequency components of heart sounds while filtering out irrelevant noise. The resulting signals then underwent a multi-stage preprocessing pipeline. Firstly, a spike removal algorithm eliminated impulsive artifacts. Secondly, a spectral subtraction technique was applied to further suppress stationary background noise and recover cleaner signals. This ensured that the resulting PCG recordings served as reliable approximations of clean heart sounds. To reduce variability in amplitude and facilitate model training, all the signals were normalized to the range $[-1, 1]$. Instead of selecting a specific part of the signal, the entire recording was utilized, ensuring that all the available heart sound information contributed to a model training, which was divided into 5 s long segments. This step was critical in maximizing the dataset's robustness and capturing diverse acoustic characteristics across the recordings.

Each 5 s long segment was further divided into non-overlapping windows of 0.8 s. The choice of 0.8 s was made to capture meaningful cardiac cycles while maintaining computational efficiency and allowing the model to process multiple shorter segments rather than a single long recording. This approach also helped to prevent overfitting to specific parts of the heart cycle.

To simulate real-world noise conditions, these preprocessed clean heart sounds were artificially contaminated using lung sounds sourced

from the ICBHI 2017 dataset. The lung sounds were preprocessed using a sixth-order Butterworth bandpass filter (cutoff frequencies of 50–2500 Hz), resampled to 1 kHz, and normalized to match the target amplitude range. Only lung sounds from the anterior right, posterior right, and lateral right auscultation sites were used, as these positions exhibit minimal overlap with cardiac signals. In each 0.8-s segment, lung sounds were added to the clean PCG recordings at five different signal-to-noise ratio (SNR) levels: −6, −3, 0, 3, and 6 dB. This mixing strategy produced a wide range of realistic noisy PCG signals, mimicking clinical scenarios where respiratory noise interferes with auscultation.

B. Testing data

The testing phase was designed to evaluate model's performance in both controlled synthetic conditions and realistic clinical environments. To create synthetic noise test sets, clean heart sounds coming from the OAHS dataset were mixed with the recordings coming from the ICBHI and HAN dataset. This resulted in two synthetic test sets: OAHS+ICBHI, simulating lung sound interference, and OAHS+HAN, incorporating environmental hospital noise. These datasets provided a controlled setting to measure the model's ability to denoise heart sounds under predefined noise conditions.

To assess a real-world performance of the proposed approach, we utilized noisy recordings coming from the PASCAL and CirCor22 datasets. These datasets contain heart sounds contaminated by various environmental and physiological noise sources recorded during the clinical auscultations. This evaluation was critical in determining how well the model generalizes to naturally occurring noise conditions that differ from those seen during the training.

To maintain consistency across the test datasets, all heart sound recordings were resampled to 1 kHz and normalized to the range $[-1, 1]$. Since the datasets have different distributions of recording lengths, each is assigned a unique standardized length. When it comes to the PASCAL dataset, only the 149 samples labeled as noisy were selected, with each recording truncated to the first 2.4 s to focus on the most relevant segments. Regarding the CirCor22 dataset, 1623 noisy samples were used for testing, each with a duration of 6 s. Recordings in the OAHS dataset were padded to 3.5 s to standardize their duration. Finally, all the recordings were segmented into 0.8-s long windows, aligning them with the training data format to ensure a consistent input structure for model evaluation.

4.3. Experimental setup

The deep learning architectures were implemented using TensorFlow and Keras, with training and evaluation conducted on an Nvidia RTX 3090 Ti GPU and an Intel(R) Xeon(R) Platinum 8352Y CPU @ 2.20 GHz (deployed just for an evaluation in the context of the ablation study). The models were optimized using MSE loss function, ensuring effective minimization of reconstruction errors during training.

An adaptive learning rate optimizer (Adam) was employed with an initial learning rate of 10^{-5} . Training was performed with a batch size of 64 over 200 epochs, incorporating an early stopping mechanism with a patience value of 5 to prevent overfitting.

All the models investigated in this study, including the LU-Net (Ali et al., 2023) and its extended variant, were trained using exactly the same experimental setup, preprocessing pipeline, and hyperparameters. This ensured a fair and consistent comparison across all the architectures and datasets.

4.4. Evaluation metrics

To comprehensively assess the effectiveness of the proposed denoising model, we employed multiple evaluation metrics that assess signal reconstruction accuracy and noise suppression efficiency. The selected metrics include true Signal-to-Noise Ratio (SNR), estimated SNR, Percent Root-Mean-Squared Difference (PRD), and Root-Mean-Squared Error (RMSE).

A. True Signal-to-Noise Ratio (True SNR)

True SNR provides an objective measure of noise suppression by quantifying the ratio of the clean heart sound signal power to the noise power. It is computed as follows:

$$SNR = 10 \log_{10} \left(\frac{P_s}{P_n} \right) = 10 \log_{10} \left(\frac{\sum_{t=1}^N x[t]^2}{\sum_{t=1}^N n[t]^2} \right) \quad (6)$$

where $x(t)$ represents the clean signal, $n(t)$ denotes the noise component, and N is the total number of samples. Higher values indicate better noise suppression.

B. Estimated Signal-to-Noise Ratio (Estimated SNR)

Since clean reference signals are often unavailable in real-world settings, we employ an estimated SNR metric tailored for heart sounds, proposed in Ali et al. (2023). This method estimates noise and signal power by segmenting the PCG signal into S1, systole, S2, and diastole regions as follows:

$$P_n = \frac{\sum_{t=1}^{N_{sys}} x_{sys}[t]^2}{N_{sys}} + \frac{\sum_{t=1}^{N_{dias}} x_{dias}[t]^2}{N_{dias}} \quad (7)$$

$$P_{ns} = \frac{\sum_{t=1}^{N_{S1}} x_{S1}[t]^2}{N_{S1}} + \frac{\sum_{t=1}^{N_{S2}} x_{S2}[t]^2}{N_{S2}} \quad (8)$$

$$P_s = P_{ns} - P_n \quad (9)$$

where P_n , P_{ns} , and P_s denote the estimated noise power, noisy signal power, and signal power, respectively. The segments $x_{sys}[t]$, $x_{dias}[t]$, $x_{S1}[t]$, and $x_{S2}[t]$ represent the systole, diastole, S1, and S2 regions of the PCG signal. Their respective lengths are given by N_{sys} , N_{dias} , N_{S1} , and N_{S2} .

C. Percent Root-Mean-Squared Difference (PRD)

PRD quantifies a deviation between the denoised and clean reference signals, evaluating signal distortion, and is defined by the following formula:

$$PRD = \sqrt{\frac{\sum_{t=1}^N (x[t] - \hat{x}[t])^2}{(x[t])^2}} \quad (10)$$

where $x[t]$ is the original clean heart sound signal, $\hat{x}[t]$ is the denoised signal, and N represents the total number of samples. A lower PRD value indicates high-fidelity signal reconstruction, preserving essential diagnostic features.

D. Root-Mean-Squared Error (RMSE)

RMSE measures the absolute reconstruction error, emphasizing larger deviations and is defined as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{t=1}^N (x[t] - \hat{x}[t])^2} \quad (11)$$

where $x[t]$ is the clean reference heart sound signal, $\hat{x}[t]$ is the denoised signal, and N denotes the total number of samples. Lower RMSE values reflect accurate signal recovery and effective noise reduction, ensuring minimal distortion of subtle heart sound characteristics.

5. Experimental results

This section is divided into two parts. In the first part of this section, a study benchmarking the proposed approach with several DNN architectures published in the context of the PCG denoising in the literature is described. The second part of this section presents an ablation study, which is supposed to show the interested reader a contribution of different building blocks deployed in the development of the proposed approach on its performance.

Table 2

Denoising performance of the three state-of-the-art DNN architectures coming from Ali et al. (2023) and the proposed model for the OAHS+ICBHI dataset. Output SNR (dB), PRD and RMSE were calculated across all the input SNR levels, ranging from −6 to 6 dB, for four models, namely the FCN, U-Net, LU-Net, and the proposed T-BiLSTM. The bold values denote the best performance for each metric.

Input SNR (dB)	Output SNR (dB)				PRD				RMSE			
	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM
−6	1.608	0.084	2.095	2.346	0.917	0.859	0.816	0.812	0.145	0.159	0.137	0.129
−3	2.910	2.724	3.984	4.157	0.869	0.803	0.752	0.748	0.126	0.129	0.111	0.107
0	4.107	4.710	5.846	5.926	0.805	0.721	0.669	0.652	0.110	0.103	0.091	0.088
3	5.042	6.423	7.447	7.324	0.730	0.620	0.577	0.580	0.099	0.085	0.077	0.079
6	5.649	7.710	8.695	8.416	0.661	0.527	0.493	0.501	0.094	0.074	0.067	0.070
Avg.	3.863	4.330	5.613	5.634	0.796	0.706	0.661	0.659	0.115	0.110	0.097	0.095

^a The results for this model were adopted from the original study (Ali et al., 2023).

Table 3

Denoising performance of the three state-of-the-art DNN architectures coming from Ali et al. (2023) and the proposed model for the OAHS+HAN dataset. Output SNR (dB), PRD and RMSE were calculated across all the input SNR levels, ranging from −6 to 6 dB, for four models, namely the FCN, U-Net, LU-Net, and the proposed T-BiLSTM. The bold values denote the best performance for each metric.

Input SNR (dB)	Output SNR (dB)				PRD				RMSE			
	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM	FCN ^a	U-Net ^a	LU-Net ^a	Proposed T-BiLSTM
−6	1.044	−1.371	1.069	1.223	0.926	0.737	0.633	0.604	0.145	0.200	0.151	0.145
−3	2.988	1.131	3.540	3.789	0.879	0.680	0.572	0.561	0.123	0.151	0.115	0.111
0	4.207	4.060	5.991	6.077	0.815	0.620	0.532	0.529	0.108	0.108	0.088	0.087
3	5.069	6.769	7.896	7.819	0.742	0.550	0.485	0.483	0.099	0.080	0.072	0.074
6	5.641	8.671	9.194	8.919	0.674	0.480	0.431	0.434	0.094	0.065	0.063	0.065
Avg.	3.789	3.851	5.537	5.565	0.807	0.613	0.530	0.522	0.114	0.121	0.098	0.096

^a The results for this model were adopted from the original study (Ali et al., 2023).

5.1. Benchmark study

This section describes experiments conducted to denoise PCG signals involving several DNNs. The primary objective of these experiments was to enhance a quality of heart sound recordings by reducing noise and improving signal clarity, which is critical for accurate diagnosis and analysis. The study explores various DNN architectures and techniques to effectively separate noise from the underlying heart sound signals, ensuring robust performance across diverse recording conditions.

Firstly, we evaluated a performance of the denoising models trained on the CinC16 dataset following the setup proposed in Ali et al. (2023) in order to fairly compare the investigated models. The models included benchmark architectures such as FCN, U-Net, LU-Net, and our proposed T-BiLSTM architecture. Table 2 compares their denoising performance for a dataset that combines relatively clean data coming from the OAHS dataset with noisy data originated in the ICBHI dataset (a lung sound data), referred to as the OAHS+ICBHI dataset. The LU-Net consistently outperforms the FCN and U-Net, achieving an average output SNR of 5.613 dB. The T-BiLSTM provides higher output SNR values at zero and the negative input SNR levels (e.g., up to an 11% increase at −6 dB) compared to the LU-Net, highlighting its improved performance in the worst-case scenarios. Additionally, the T-BiLSTM model achieves the highest average output SNR of 5.634 dB.

The performance comparison between the original (relatively clean) and denoised output signals shows similar behavior to that observed with the SNR metric. A lower value when it comes to the PRD and RMSE metrics, indicates less deviation between the signals, thus implying better reconstruction. There is a consistent decrease in the PRD and RMSE values as an input SNR increases for all the investigated models. At the low input SNR levels, the FCN outperforms the U-Net in terms of the output SNR and RMSE values. This suggests that the latent representation learned by the FCN-based denoising autoencoder may be more effective in capturing relevant patterns in noisy signals. One possible explanation is that the FCN, by design, focuses on extracting and reconstructing essential features without the influence of skip connections, which might sometimes reintroduce noise in the U-Net approach.

The LU-Net achieves lower PRD and RMSE values than the FCN and U-Net architectures, on average 0.661 and 0.097, respectively. Meanwhile, the T-BiLSTM performs slightly better, with the average PRD of 0.659 and RMSE of 0.095. These improvements are particularly noticeable at zero and the negative input SNR levels. For instance, the most discernible enhancement is a 5.83% reduction obtained for the RMSE metric at −6 dB, along with the improvements of 3.60% and 3.30% at −3 dB and 0 dB, respectively.

Table 3 compares a performance of the denoising models tested on a dataset combining relatively clean data coming from the OAHS dataset and noisy data originated in HAN dataset (a hospital noise), referred to as the OAHS+HAN dataset. On the basis of the reported values of the output SNR, PRD, and RMSE metrics, we can say that the U-Net achieves similar accuracy in a signal reconstruction at 3 dB and 6 dB input levels in comparison to the LU-Net. However, the T-BiLSTM outperforms the LU-Net at 0 dB, −3 dB, and −6 dB levels, highlighting the benefits of the proposed architectural improvements.

Compared to the performance of the models in the presence of the lung noise, the hospital noise had a negative impact on a signal reconstruction in terms of the output SNR at the low input levels (−3 dB and −6 dB). However, at the higher input levels (3 dB and 6 dB), the hospital noise led to an improved performance. This pattern is consistently observed for both U-Net-based and T-BiLSTM architectures, where the RMSE values degraded at the low input levels for the hospital noise but improved at the higher input levels. This behavior might be caused by the characteristics of the hospital noise. At the low input SNR levels, the noise could be masking important signal features, making it harder for the model to extract relevant information. Conversely, at the higher input SNR levels, the model may be able to leverage the structured nature of the hospital noise for enhancing its denoising capability, possibly by learning to distinguish it more effectively from the underlying signal.

For the subsequent experiments, the best performing models in the previous phase, i.e., the LU-Net (Ali et al., 2023) and T-BiLSTM, were selected for training on a significantly larger and more diverse dataset, resulting in the LU-Net* and the T-BiLSTM* models. This dataset combines multiple sources, including the CDHS, CinC16, CirCor22, HCT-11,

Table 4

Denosing performance of the previously best performing state-of-the-art DNN architectures (LU-Net (Ali et al., 2023) and T-BiLSTM) further retrained on the large and more diverse dataset forming the LU-Net* and the proposed T-BiLSTM* models for the OAHS+ICBHI dataset. Output SNR, PRD and RMSE were calculated over a range of the input SNR levels (−6 to 6 dB). The bold values denote the best performance for each metric.

Input	Output SNR (dB)		PRD		RMSE	
SNR (dB)	LU-Net*	Proposed T-BiLSTM*	LU-Net*	Proposed T-BiLSTM*	LU-Net*	Proposed T-BiLSTM*
−6	2.435	2.582	0.789	0.785	0.129	0.124
−3	4.235	4.478	0.723	0.717	0.104	0.101
0	6.125	6.346	0.634	0.621	0.087	0.083
3	7.533	7.652	0.576	0.568	0.074	0.075
6	8.512	8.541	0.485	0.487	0.066	0.066
Avg.	5.768	5.919	0.641	0.635	0.092	0.089

Table 5

Denosing performance of the previously best performing state-of-the-art DNN architectures (LU-Net (Ali et al., 2023) and T-BiLSTM) further retrained on the large and more diverse dataset forming the LU-Net* and the proposed T-BiLSTM* models for the OAHS+HAN dataset. Output SNR, PRD and RMSE were calculated over a range of the input SNR levels (−6 to 6 dB). The bold values denote the best performance for each metric.

Input	Output SNR (dB)		PRD		RMSE	
SNR (dB)	LU-Net*	Proposed T-BiLSTM*	LU-Net*	Proposed T-BiLSTM*	LU-Net*	Proposed T-BiLSTM*
−6	1.188	1.395	0.622	0.596	0.147	0.139
−3	3.670	3.952	0.561	0.548	0.110	0.107
0	6.051	6.142	0.524	0.519	0.086	0.084
3	8.005	8.093	0.479	0.478	0.071	0.069
6	9.225	9.278	0.428	0.426	0.062	0.062
Avg.	5.628	5.772	0.523	0.513	0.095	0.092

and ZCHSound datasets, to cover a broader range of heart sound recordings. Since these datasets were collected under varying microphone settings, it is expected that they will be incorporated during a training process into a newly trained models. It is worth noting here that the corresponding incorporation is supposed to enhance a robustness and generalizability of the investigated denoising architectures. Moreover, it should be mentioned here that the LU-Net* model was trained using exactly the same dataset, preprocessing steps, loss function, optimizer, and training configuration as the proposed T-BiLSTM* model, ensuring a fair comparison between both approaches.

The outcomes of the experiments involving two different combinations of the datasets, i.e., OAHS+ICBHI and OAHS+HAN, are summarized in Tables 4 and 5, respectively. For both datasets, the T-BiLSTM* generally outperforms the LU-Net* in terms of the average output SNR, PRD and RMSE. Moreover, the improvements were more pronounced at the input SNR levels ranging from 0 dB to −6 dB, indicating that the proposed model is particularly effective at enhancing the severely degraded signals.

The substantial expansion of the training dataset resulted in a notable accuracy enhancement for the T-BiLSTM, particularly in the presence of the lung noise (the OAHS+ICBHI dataset). The model achieved a 5.06% increase in the average output SNR, a 3.64% improvement in the average PRD, and a 6.32% reduction in the average RMSE (all calculated as percentage changes). However, when it comes to the OAHS+HAN dataset covering the hospital noise, the enhancement was less pronounced, with a 3.72% increase in the output SNR, a 1.72% improvement in the PRD, and a 4.17% reduction in the RMSE.

It should be highlighted here that the LU-Net benefited less from the dataset expansion compared to the T-BiLSTM. For instance, when it comes to the OAHS+ICBHI dataset, the LU-Net showed only a 2.76% increase in the average output SNR, a 3.03% reduction in the average PRD, and a 5.15% decrease in the average RMSE. This suggests that the T-BiLSTM, which embeds a transformer within the U-Net, benefits more from the larger and more diverse dataset. It is worth noting here that transformers excel at capturing long-range dependencies and complex patterns within data, which may allow the T-BiLSTM to generalize better as the dataset grows.

The models were initially tested on artificially created noisy heart sound recordings, using two different combinations of datasets, i.e.,

OAHS+ICBHI and OAHS+HAN, as reported above. To assess their denoising capabilities in real-world clinical environments, datasets such as the PASCAL and CirCor22 were used. The PASCAL dataset includes noisy heart sound data coming from various clinical and non-clinical environments. The CirCor22 dataset contains pediatric heart sound data with diverse background noises like stethoscope rubbing, speech, crying and laughter. The models' denoising performance was evaluated by deploying the estimated SNR metric, with the results presented in Table 6 for the PASCAL dataset and Table 7 for the CirCor22 dataset, respectively. The models trained on the extended PCG dataset (LU-Net* and T-BiLSTM*) outperformed, as clearly expected, those trained solely on the CinC16 dataset used in the initial experiments. Notably, the proposed T-BiLSTM* architecture achieved the highest estimated SNR of 14.984 dB for the PASCAL dataset and 11.924 dB for the CirCor22 dataset. We believe that pre-training on a larger and more diverse dataset significantly contributes to the model's robustness and generalizability. Our approach incorporates five distinct and sizable PCG datasets from different sources, enhancing the model's ability to generalize effectively across varying noise types, recording conditions and clinical environments.

5.2. Ablation study

This section presents an ablation study that evaluates the contribution of each component to the proposed T-BiLSTM model. The study was designed to assess whether an inclusion of the BiLSTM layers and transformer blocks contributes to an improved denoising performance, and if so, to what extent. Additionally, an effect of deploying the extended training dataset was examined. All the experiments were performed using a fixed input SNR of 6 dB. This value was selected because denoising becomes more difficult at higher SNR levels, where the noise is less dominant and thus harder to distinguish from a clean heart sound signal. The analysis is based on two combinations of the datasets, namely OAHS+ICBHI and OAHS+HAN. The results of the ablation study for both the above-mentioned combinations of the datasets are displayed in Tables 8 and 9, respectively.

For the OAHS+ICBHI dataset, the base U-Net model achieved an output SNR of 7.782 dB when trained on the extended dataset. When

Table 6

Denoising performance for the PASCAL dataset covering the real-world noise. The bold values denote the best performance for each metric. The models without an asterisk (*) were trained solely on the CinC16 dataset, while the models marked with an asterisk (*) were trained on a composite dataset comprising multiple sources, i.e., CDHS, CinC16, CirCor22, HCT-11 and ZCHSound datasets.

Performance metric	Noisy signal	Denoised signal				
		FCN	U-Net	LU-Net	LU-Net*	Proposed T-BiLSTM*
Estimated SNR (dB)	7.456	13.635	10.348	14.017	14.461	14.984

Table 7

Denoising performance for the CirCor22 test dataset covering the real-world noise. The bold values denote the best performance for each metric. The models without an asterisk (*) were trained solely on the CinC16 dataset, while the models marked with an asterisk (*) were trained on a composite dataset comprising multiple sources, i.e., CDHS, CinC16, CirCor22, HCT-11 and ZCHSound datasets.

Performance metric	Noisy signal	Denoised signal				
		FCN	U-Net	LU-Net	LU-Net*	Proposed T-BiLSTM*
Estimated SNR (dB)	5.749	8.563	9.124	10.862	11.587	11.924

only the BiLSTM layers were added, the SNR increased to 8.198 dB, representing an improvement of +0.416 dB. Incorporating only the transformer block yielded an SNR of 8.342 dB, which is a larger gain of +0.560 dB over the U-Net baseline. The proposed T-BiLSTM model, which combines both the BiLSTM and transformer components, achieved the highest SNR of 8.541 dB, corresponding to a total improvement of +0.759 dB over the base model. Similar improvements were also observed in terms of the PRD and RMSE metrics. It is worth noting here that the lower performance for all the deployed metrics was obtained for the original training dataset, i.e., CinC16. For the OAHS+HAN dataset, the base U-Net achieved an SNR of 8.693 dB with the extended training data. Adding only the BiLSTM layers increased the SNR to 8.829 dB, providing a modest gain of +0.136 dB. The model with only the transformer block reached 8.961 dB, improving over the U-Net baseline by +0.268 dB. The full T-BiLSTM model achieved an SNR of 9.278 dB, representing a total gain of +0.585 dB. As already reported above for the OAHS+ICBHI dataset combination, the improvements in terms of the PRD and RMSE followed the same pattern, confirming the consistent benefit of using both architectural components. Similarly as already reported above for the OAHS+ICBHI dataset combination, the lower performance for all the deployed metrics was achieved for the original training dataset. The results confirm that the transformer block has a stronger individual effect than BiLSTM and that the best results are achieved when both components are used together.

To evaluate the effect of the transformer component under different training conditions, we compared its performance for the models trained on the original and extended dataset. For the OAHS+ICBHI dataset, adding the transformer to the U-Net architecture improved the SNR from 7.710 dB to 8.191 dB for the model trained on the original dataset (+0.481 dB), and from 7.782 dB to 8.342 dB for the model trained on the extended dataset (+0.560 dB). Similarly, for the OAHS+HAN dataset, the transformer improved the SNR from 8.671 dB to 8.836 dB for the model trained on the original dataset (+0.165 dB), and from 8.693 dB to 8.961 dB for the model trained on the extended dataset (+0.268 dB). These results demonstrate that while the transformer enhances the performance in both scenarios, the gain is greater when more diverse data is available. This indicates that the transformer block benefits more from the data scale and variety, compared to the other components, as clearly expected.

In addition to the denoising performance, we evaluated the computational cost of each model. The inference time was calculated as the average time required to process a 0.8-s signal segment across the test set. On the GPU, the U-Net model achieved an inference time of 0.70 ms, while the proposed T-BiLSTM model required 2.55 ms. On

the CPU, the inference time increased to 2.80 ms for the U-Net and to 17.65 ms for the T-BiLSTM. The number of the parameters also grew from 1.256 million (M) for the U-Net to 1.736 M for the T-BiLSTM. Although the proposed model is more complex, its execution time remains within the range acceptable for a real-time processing, considering the fact that the real-time applications, when it comes to the biomedical signal processing, typically allow latencies up to 100 to 300 ms.

The findings presented in this section demonstrate that the improvements are not only the result of the larger dataset. The transformer block makes a clear and consistent contribution to a model performance. Across both datasets, it provides a greater benefit than the BiLSTM when used independently. This supports the conclusion that the transformer component is an important building block of the proposed architecture for achieving the superior performance of the proposed T-BiLSTM model.

6. Discussion

This study investigates PCG signals denoising using various DNN architectures to enhance a signal quality for improved diagnostic accuracy. Models such as the FCN, U-Net, LU-Net, and the proposed T-BiLSTM were initially tested using relatively clean data mixed with noisy data.

Firstly, the models were trained on the CinC16 dataset and evaluated on the OAHS+ICBHI dataset containing the lung noise. As expected, all the models demonstrated a consistent increase in the output SNR as well as a decrease in the PRD and RMSE values as an input SNR increased. The LU-Net outperformed the FCN and U-Net, achieving an average output SNR of 5.613 dB. The T-BiLSTM achieved the highest average output SNR of 5.634 dB, particularly excelling at zero and the negative input SNR levels (up to an 11% improvement at -6 dB compared to the LU-Net). The T-BiLSTM also obtained the lowest average PRD (0.659) and RMSE (0.095), with noticeable improvements at zero and the negative input SNR levels. In some instances, the FCN performed better than the U-Net at the low input SNRs, likely due to its architecture focusing on essential feature extraction without skip connections, which may reintroduce noise in the U-Net.

In the experiments combining the OAHS data with the noisy data coming from the HAN dataset (the hospital noise), the T-BiLSTM showed similar accuracy to the LU-Net at the higher input levels (3 dB and 6 dB) but outperformed the LU-Net at the lower input levels (0 dB, -3 dB, and -6 dB). Compared to the lung noise, the hospital noise negatively impacted the output SNR at the low input levels but improved the performance at the higher input levels. This trend was

Table 8

Ablation study evaluating the impact of the BiLSTM and transformer components, as well as the use of the extended training dataset, on the performance of the proposed T-BiLSTM model. Performance is reported for the OAHS+HAN dataset at an input SNR of 6 dB by applying the standard denoising metrics (SNR, PRD, RMSE). The table also includes a model size and inference times obtained on the GPU and CPU. Bold values indicate the best performance for the corresponding metric.

Model variant	Original training dataset	Extended training dataset	SNR [dB]	PRD	RMSE	Param. [M]	Infer. time GPU [ms]	Infer. time CPU [ms]
Base U-Net	✓		7.710	0.527	0.074	1.256	0.70	2.80
Base U-Net		✓	7.782	0.523	0.073	1.256	0.70	2.80
+ BiLSTM	✓		8.013	0.508	0.071	1.320	1.29	4.65
+ BiLSTM		✓	8.198	0.499	0.069	1.320	1.29	4.65
+ Transformer	✓		8.191	0.498	0.070	1.672	2.07	15.13
+ Transformer		✓	8.342	0.494	0.069	1.672	2.07	15.13
+ BiLSTM + Transformer	✓		8.456	0.492	0.068	1.736	2.55	17.65
+ BiLSTM + Transformer (proposed)		✓	8.541	0.487	0.066	1.736	2.55	17.65

Table 9

Ablation study evaluating the impact of the BiLSTM and transformer components, as well as the use of the extended training dataset, on the performance of the proposed T-BiLSTM model. Performance is reported for the OAHS+HAN dataset at an input SNR of 6 dB by applying the standard denoising metrics (SNR, PRD, RMSE). The table also includes a model size and inference times obtained on the GPU and CPU. Bold values indicate the best performance for the corresponding metric.

Model variant	Original training dataset	Extended training dataset	SNR [dB]	PRD	RMSE	Param. [M]	Infer. time GPU [ms]	Infer. time CPU [ms]
Base U-Net	✓		8.671	0.480	0.065	1.256	0.68	2.77
Base U-Net		✓	8.693	0.476	0.065	1.256	0.68	2.77
+ BiLSTM	✓		8.748	0.470	0.064	1.320	1.27	4.61
+ BiLSTM		✓	8.829	0.461	0.063	1.320	1.27	4.61
+ Transformer	✓		8.836	0.461	0.063	1.672	2.03	15.10
+ Transformer		✓	8.961	0.455	0.062	1.672	2.03	15.10
+ BiLSTM + Transformer	✓		8.919	0.434	0.064	1.736	2.51	17.42
+ BiLSTM + Transformer (proposed)		✓	9.278	0.426	0.062	1.736	2.51	17.42

consistent across all the U-Net-based models. The difference in the performance can be attributed to the structured nature of the hospital noise, which can obscure the signal features at the low input levels but may not pose a challenge for denoising at the higher input levels.

The most effective/best performing models in the previous phase, i.e., LU-Net and T-BiLSTM, were selected for further experiments using the larger and more diverse training dataset combining multiple heart sound sources, namely CDHS, CinC16, CirCor22, HCT-11, and ZCHSound datasets, to improve the robustness and generalizability of the investigated models, resulting in the LU-Net* and the T-BiLSTM* models.

The T-BiLSTM* demonstrated notable advancements in an accuracy with the expanded dataset, particularly for the OAHS+ICBHI dataset. A 5.06% increase in the output SNR, a 3.64% improvement in the PRD, and a 6.32% reduction in the RMSE were reported in this context. When it comes to the OAHS+HAN dataset, smaller gains were achieved with a 3.72% increase in the output SNR, a 1.72% improvement in the PRD, and a 4.17% reduction in the RMSE. The difference in the accuracy gains can be attributed to the fact that the model was trained on the heart sound recordings containing the lung noise, indicating that it learned to handle this type of noise more effectively.

To sum up, while both architectures perform rather well, the T-BiLSTM* provides slightly better denoising performance across all the metrics for both datasets, especially at the input SNR levels ranging from 0 dB to −6 dB, effectively improving the severely degraded signals. Despite the fact that all the architectures benefit from the increased data volume, the larger dataset allowed the transformer-based T-BiLSTM* model to better leverage its ability to capture long-range dependencies and complex temporal patterns, leading to improved generalization.

Lastly, considering the challenges posed by real-world environments, the performance evaluation of the final models, i.e., the T-BiLSTM* and LU-Net*, was also conducted for the PASCAL and CirCor22 datasets. It is worth reiterating here that these datasets include natural ambient noise that occurs during the recording of heart sounds,

allowing a more realistic assessment of the models' performance. Overall, the T-BiLSTM* achieved the best denoising results for both datasets. The findings presented in this paper highlight an importance of robust model architecture for effectively separating noise from heart sound signals and underscore a challenge induced by a presence of different noise types in this context, especially those originating in medical environments. The proposed model's ability to handle such complex noise scenarios demonstrates its enhanced robustness and generalizability, which are critical for practical deployment in clinical settings.

In the ablation study, we systematically analyzed the contribution of each architectural component to the proposed T-BiLSTM model. The results obtained for two different combinations of the datasets, i.e., OAHS+ICBHI and OAHS+HAN, consistently showed that both the BiLSTM layers and transformer blocks improve the denoising performance in comparison to the baseline U-Net model. The transformer block alone provided a larger performance gain than the BiLSTM when added individually, and the best results were obtained when both components were applied together. These improvements were confirmed across both the original and extended training datasets, with the transformer benefiting more from the availability of the additional training data as clearly expected. Although the enhanced model increased computational cost, the inference times remained well below the threshold required for the real-time biomedical signal processing tasks. These findings demonstrate that the performance gains are not solely due to the data volume but they result directly from the proposed architectural enhancements.

7. Conclusion

In this work, we proposed the novel deep convolutional architecture entitled the T-BiLSTM that integrates the BiLSTM layers with the transformer blocks involving multi-head attention mechanism, which was particularly designed for denoising PCG signals. The extensive benchmark of the proposed T-BiLSTM architecture with the state-of-the-art denoising approaches involving both the synthetic data as

well as the real-world clinical conditions demonstrated the significant improvements in terms of all the evaluated signal quality and noise suppression metrics as well as for all the used benchmark datasets, in comparison to the state-of-the-art approaches. Moreover, the comparative analysis underscores the effectiveness of the proposed model in preserving the critical heart sound characteristics while eliminating ambient and physiological noise. It is worth noting here that the real-world environment evaluation presented in this study is more complex, involving two different and renowned testing datasets, than those done in the previous studies, e.g., Ali et al. (2023), Al-Zaben et al. (2024). So, we believe that this ensures a higher robustness and generalizability of the proposed approach.

The ablation study confirmed that both the BiLSTM and transformer components contribute to the improved performance of the proposed model, with the transformer providing a greater individual benefit. The highest denoising performance was achieved when both components were applied together. While a higher complexity increases an inference time of the proposed model, the model still remains suitable for a deployment in resource-constrained environments, as it operates well below the latency limits required for the real-time biomedical signal processing applications/tasks.

Future work will focus on enhancing the proposed denoising model by exploring advanced preprocessing techniques, such as adaptive filtering and dynamic noise suppression, to improve an input signal quality. Further architectural optimizations could include hybrid models that integrate novel convolutional, recurrent, and attention-based mechanism to capture both spatial and temporal dependencies more effectively. Introducing domain adaptation strategies, such as a transfer learning and Generative Adversarial Networks, could further improve the model robustness in diverse clinical environments.

CRediT authorship contribution statement

Maros Jakubec: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Eva Lieskovska:** Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Peter Pocta:** Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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